

Highlights

Detection Quality Index (DQI): Inverse Optimization Methodology Based on Genetic Algorithms for Damage Identification in Jacket-Type Offshore Platforms

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- A Detection Quality Index (DQI) fusing eight vibration indicators with exponential false-positive penalization improves damage detection reliability in noisy offshore environments.
- Secondary elements (braces) behave as structural fuses, detectable from 30% severity, achieving 92.2% (denting) and 96.5% (corrosion) global detection rates.
- Main elements (legs) require damage severity exceeding 50% for reliable detection, indicating strong structural redundancy.
- A Topology-Weighted Precision metric corrects classical Precision underestimation by up to +22 percentage points for brace elements in jacket lattice structures.
- Validated over 3,240 simulation runs on a calibrated finite-element model under corrosion and denting scenarios with AUC-ROC > 0.90 for all element types.

Detection Quality Index (DQI): Inverse Optimization Methodology Based on Genetic Algorithms for Damage Identification in Jacket-Type Offshore Platforms

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ARTICLE INFO

Keywords:

Structural Health Monitoring
Genetic Algorithms
Jacket Platforms
Detection Quality Index
Inverse Optimization
Topology-Weighted Precision

ABSTRACT

The aging of offshore oil infrastructure and the high costs and risks associated with traditional underwater inspections demand new integrity management strategies. This paper proposes a Structural Health Monitoring (SHM) methodology based on inverse optimization using Genetic Algorithms (GA) for damage identification in Jacket-type platforms. The Detection Quality Index (DQI) is introduced as a new objective metric designed to fuse multiple vibration indicators and mitigate numerical uncertainty by penalizing false positives. Validation was conducted over 3,240 simulation runs on a calibrated finite-element model under corrosion and denting scenarios. Quantitative results demonstrate a differentiated structural behavior: secondary elements (braces) act as “structural fuses”, detectable from 30% damage severity with global detection rates of 92.2% (denting) and 96.5% (corrosion), while main elements (legs) require severity levels exceeding 50%. A novel Topology-Weighted Precision metric is also proposed, correcting systematic underestimation of localization quality by the classical Precision metric, yielding gains of up to +22 percentage points for brace elements.

1. Introduction

The global energy transition and the mounting pressure on offshore asset operators to extend the service life of their installations have elevated structural integrity management to a strategic priority of the highest order (Hlaing, 2024). Jacket-type platforms, characterised by their three-dimensional steel lattice configuration, form the backbone of hydrocarbon production facilities in shallow and intermediate waters, and are playing an increasingly important role as foundations for offshore wind turbines (Weijters, Verbeelen, Capaldo and Devriendt, 2024). These structures must simultaneously withstand production deck loads, stochastic hydrodynamic forces from waves and currents, and, in the case of wind turbines, the inertial forces transmitted from the nacelle and blades. In strategic regions such as the Gulf of Mexico, a significant proportion of fixed infrastructure has exceeded its original design service life—typically 20 to 25 years—and continues to operate under conditions of accelerated corrosion-induced and fatigue-driven deterioration (Mubarak, Ahmad and Kamaruddin, 2020). Historically, the integrity management of these structures has relied on periodic visual inspections conducted by industrial divers or remotely operated vehicles (ROVs), methods that not only entail exorbitant operational costs and significant personnel risks (Instituto Mexicano del Petróleo, 2020), but are systematically inadequate for detecting internal or localised damage at early stages—a limitation further compounded by sector-wide maintenance budget cuts.

2. Literature Review

The marine environment subjects jacket structures to a convergence of degradation mechanisms of stochastic and electrochemical nature that simultaneously compromise their strength and dynamic response. Corrosion, exacerbated by salinity, humidity, and seawater temperature variations, does not act uniformly along the structure: integrity studies in tropical environments document corrosion rates in the Splash Zone of between 0.28 and 0.35 mm/year—values 140% to 175% higher than those assumed in the original design—with localised pitting of up to 4.8 mm recorded over as little as 18 months of operation (Al-Yacoubi, Khaleel and Liew, 2020). This cross-sectional loss raises the Unity Check (UC) utilisation ratio—the ratio of acting stress to member resistance capacity—whose critical operational threshold is $UC = 0.8$ in accordance with current structural integrity guidelines (American Petroleum Institute, 2014; Salama, Liew and Al-Yacoubi, 2020). Simultaneously, marine growth accumulation on tubular members increases their effective hydrodynamic diameter, amplifying drag forces according to the Morison equation and shifting the natural frequencies of the structure in a way that can mask modal changes caused by stiffness loss (Salama et al., 2020). These mechanisms are compounded by fatigue cracking, driven by the cyclic nature of wave loading and operational vibrations, which initiates preferentially at the weld toe of tubular joints in T, K, and Y configurations—high stress concentration points—and propagates silently with globally imperceptible stiffness changes for distributed sensor networks (Raza, Zhao, Zhang and Umar, 2021).

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In response to these limitations, Structural Health Monitoring (SHM) has undergone a paradigm shift between 2018 and 2026, migrating from periodic inspections towards real-time diagnostic systems based on vibratory response and artificial intelligence (Ghsoub and Siriwardane, 2018; Weijtens et al., 2024). Vibration-based damage identification exploits the physical relationship between modal properties—natural frequencies and mode shapes—and the structural stiffness state: a local stiffness reduction modifies global modal properties in a way that, in principle, can be inverted to localise and quantify damage. This approach is formalised as a model updating problem: given experimental modal parameters, the finite-element model parameters are calibrated to minimise the discrepancy between simulated and measured responses (Doebling, Farrar, Prime and Shvitz, 1996). However, the solution to this inverse problem is inherently ill-conditioned—non-unique and sensitive to perturbations—when attempting to parametrically adjust a structure with hundreds of members from a limited number of experimentally identified modes, frequently leading to physically inconsistent solutions (Aghaeidoost and Sharbatdar, 2023).

The most advanced model updating methodologies, proposed between 2022 and 2024, address this high-dimensionality problem through sensitivity-based clustering, which categorises structural members into homogeneous groups of dynamic behaviour to reduce the effective number of parameters to be optimised (Jiang, Yan and Soares, 2024). Natural frequency error and modal shape discrepancy are simultaneously adopted as objective functions, with the latter evaluated through the Modal Assurance Criterion (MAC), whose unit value indicates perfect vector correlation between the experimental and numerical eigenvectors (Jiang et al., 2024). Solving both error functions concurrently constitutes a multi-objective optimisation problem for which gradient-based methods are unsuitable given the model non-linearity and multiplicity of local minima.

In this context, Genetic Algorithms (GAs) have consolidated their position as the reference search engine for inverse optimisation in offshore SHM (Goldberg, 1989). Their principal advantage lies in the capacity to explore high-dimensional solution spaces without requiring gradient information, while maintaining notable robustness against local minima and noisy input data. Previous research on jacket platforms has demonstrated that GAs can identify damage location and severity with estimation errors of 3% to 5% when modal data are contaminated with random noise of up to 3%, validating their applicability under realistic offshore measurement conditions (Rahmani and Behzad, 2013). Nevertheless, the effectiveness of these algorithms depends critically on the objective function formulation: a function that fuses multiple damage indicators and effectively penalises false positives constitutes the key differentiating element for reliable diagnostic systems in offshore operations (Malekzadeh, Atia and Catbas, 2013).

An aspect that has received renewed attention in the period 2022–2026 is the quantitative assessment of damage

localisation quality. The classical Precision metric penalises all false positives (FPs) identically, regardless of their spatial proximity to the truly damaged element, which is inadequate in lattice structural networks where an adjacent FP (topological distance $d = 1$) carries a radically different engineering interpretation from one at the opposite end of the structure. Recent work has proposed distance-weighted metrics—such as MASDWP (*Mean Average Spatial Distance-Weighted Precision*)—that provide a more honest assessment of localiser performance in large-scale structures (Li and Zou, 2024). In parallel, the Spatio-Temporal Graph Machine Learning (TVML) framework characterises the sensor network through centrality metrics—*TopoScore*—selecting representative nodes from each modal cluster to reduce instrumentation cost without sacrificing diagnostic fidelity (Ma, Yan and Shen, 2024).

The quantitative validation of these systems relies on standard classification metrics—Precision, Recall, and the area under the ROC curve (AUC-ROC)—which enable evaluation of the trade-off between true positive rate and false alarm cost independently of the decision threshold (Li and Zou, 2024). Despite this scientific maturity, a significant normative gap persists: current international standards—ISO 19902, API RP 2S1M, DNV-RP-C210, and NORSOK N-005—provide general integrity management guidelines without digital twins and automated damage detection having attained regulatory requirement status (Petroleum Safety Authority Norway, 2022).

The work most closely related to the present study is that of Ervin and Zeng (Ervin and Zeng, 2022), who proposed combining multiple DIs through a fusion metric and demonstrated that the optimal combination outperforms any individual indicator, achieving detection rates of 94% for 60% reductions in Young's modulus under 5% noise conditions. However, their formulation employed a binary target vector and was validated on planar steel frames, without incorporating false-positive penalisation or addressing the hydrodynamic complexities, structural redundancy, and diversity of damage mechanisms (corrosion + denting) characteristic of offshore jacket platforms.

The state-of-the-art review thus reveals three technical gaps motivating the present research. First: the absence of a unified objective function that fuses multiple vibratory indicators and exponentially penalises false positives, adapted to the noise conditions inherent to platforms in operation. Second: the lack of a localisation metric that corrects the systematic underestimation of diagnostic quality caused by classical Precision in structurally redundant lattice networks. Third: the absence of exhaustive statistical validation of the operational hypothesis positing differentiated behaviour between secondary elements (braces or diagonals) and primary elements (legs) of the lattice under progressive corrosion and denting damage.

This study proposes an Inverse Optimisation strategy based on Genetic Algorithms for damage identification in jacket platforms. The central contribution is the **Detection Quality Index (DQI)**, a novel objective metric designed to

fuse multiple damage indicators and exponentially penalise false positives, improving diagnostic reliability in noisy operational environments (Malekzadeh et al., 2013). As a second methodological contribution, the **Topology-Weighted Precision** ($Precision_w$) is proposed, which assigns a weight $w(d) = d/(d+1)$ to each false positive as a function of its distance to the damaged element in the structural connectivity graph—a correction that, as demonstrated in Section 5, yields gains of up to +22 percentage points in localiser evaluation for braces. Validation was carried out over 3,240 algorithm runs on a calibrated finite-element model of a 4-leg jacket platform subjected to corrosion (5%–90% severity) and denting (5%–45%) scenarios, with quantitative analysis of Precision, Recall, and noise robustness via AUC-ROC (Section 5.7).

The paper is organised as follows: Section 3 describes the numerical modelling of the platform, incorporating added-mass and marine growth effects. Section 4 details the GA-based methodology and the mathematical formulation of the DQI. Section 5 presents the results and discussion for the corrosion and denting scenarios, evaluating the sensitivity of the method. Finally, Section 6 offers conclusions and future research directions.

3. Numerical Modelling and Damage Formulation

The dynamic behaviour of the offshore platform is modelled by discretising the continuous structure into a multi-degree-of-freedom (MDOF) system. Under the assumption of linear elastic behaviour and small oscillations, the dynamic equilibrium is governed by the differential equation of motion for undamped free vibration:

$$\mathbf{M}\ddot{\mathbf{u}}(t) + \mathbf{K}\mathbf{u}(t) = \mathbf{0} \quad (1)$$

where $\mathbf{K}$ is the global stiffness matrix and $\mathbf{M}$ is the global mass matrix of the system ($n \times n$). The latter is critical in offshore structures, as it must incorporate not only the structural steel mass but also the complex hydrodynamic interactions described below.

3.1. Components of the Global Mass Matrix

To faithfully represent the physics of an in-service jacket platform, the mass matrix is assembled as the superposition of three components: structural mass (M_s), added mass (M_a), and trapped mass (M_t).

3.1.1. Hydrodynamic Mass and Coefficients (C_m)

The added mass represents the volume of fluid that oscillates in concert with the tubular member and is modelled through the mass coefficient C_m , which depends on surface roughness and biological fouling. In accordance with design practice and technical reports (American Petroleum Institute, 2014; Cárdenas-Arias and Maldonado-Zepeda, 2019), reference values of $C_m = 1.6$ for surfaces with significant marine growth and $C_m = 1.2$ for clean surfaces are adopted

in this study. Additionally, the trapped mass (water enclosed within flooded tubular cross-sections) is incorporated as internal added mass in the model assembly; experimental details and schematic descriptions of the definition and parametrisation of both terms are given in (Cárdenas-Arias and Maldonado-Zepeda, 2019).



3.1.2. Marine Growth Modelling

Marine growth modifies the effective mass and hydrodynamics of tubular members through variations in biological thickness along the water column; its impact depends on light exposure and depth (Al-Yacoubi et al., 2020; Salama et al., 2020). In the numerical model, a variable thickness profile was implemented by vertical zones (tidal zone vs. seabed), locally adjusting the mass and inertia coefficients in the most affected regions.

3.2. Mechanics of Simulated Damage

In contrast to black-box models, this research employs a physics-based approach to simulate damage. The geometric properties of the finite-element cross-sections are modified and the global matrices are re-assembled to obtain the updated modal properties (forward problem).

3.2.1. Uniform Corrosion

Corrosion is modelled as a uniform reduction of the wall thickness (t) of the tubular member over an affected length (L_D). This geometric loss reduces the cross-sectional area (A) and the moments of inertia (I), directly altering member stiffness. Regarding the elastic modulus (E), experimental studies by Cárdenas-Arias and Maldonado-Zepeda (2019) show that, in the linear elastic range, steel corroded by marine immersion does not exhibit significant changes in E ; consequently, E is held constant in this work and only geometric modifications (A , I) are considered for damage characterisation (Cárdenas-Arias and Maldonado-Zepeda, 2019).

3.2.2. Denting

Denting represents critical localised damage caused by accidental impacts. This damage is modelled by assuming a cross-sectional distortion that flattens the impacted face and widens the lateral surfaces, conserving the perimeter but drastically reducing the local moment of inertia (I_{red}) (Figure 1).

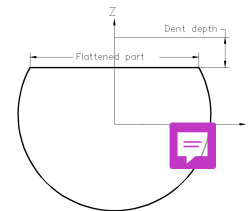


Figure 1: Parametrisation of the dented cross-section.

To avoid abrupt numerical discontinuities in the stiffness matrix, the variation of inertia along the dented zone is

smoothed through fourth-degree polynomial interpolation functions, ensuring a representative transition of local stiffness loss (Figure 2).

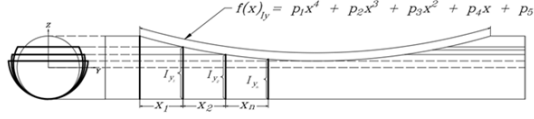


Figure 2: Longitudinal interpolation of moment of inertia in the dented zone for numerical smoothing.

3.3. Case Study: Jacket-Type Platform

The methodology is validated on a numerical model of a fixed 4-leg jacket-type offshore platform representative of structures installed in the shallow and intermediate waters of the Gulf of Mexico (Figure 3). The finite-element model incorporates the complete superstructure and the pile foundation system (simulated via equivalent length to the apparent fixity point).

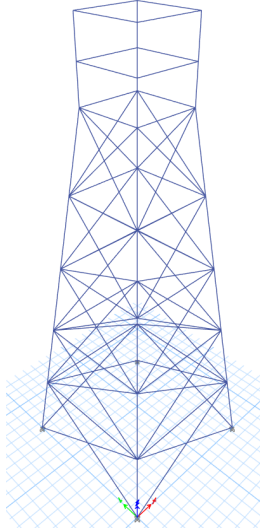


Figure 3: Numerical model of the study platform: 3D isometric view.

4. Inverse Optimisation Methodology

To address the structural damage identification problem in offshore platforms, this study extends the GA-based index combination framework proposed by Ervin and Zeng (Ervin and Zeng, 2022), who demonstrated that a GA can optimally select and weight multiple individual damage indices (DIs), outperforming any single indicator. In the present work, that framework is adapted to the offshore jacket context and an original contribution is incorporated as the objective function: the Detection Quality Index (DQI), designed to exponentially penalise false positives and maximise diagnostic reliability in noisy environments—aspects not addressed in the original formulation of Ervin and Zeng (Ervin and Zeng, 2022).

The methodology is fundamentally structured in two coupled stages: the mechanical simulation of damage scenarios and the stochastic optimisation of detection. The general operational workflow is detailed in Figure 4. That figure illustrates the complete chain from data acquisition on the real platform to maintenance decision-making; the scope of the present work is confined to the central stage highlighted in the figure: numerical damage model generation, computation of the eight DIs, and DQI+GA optimisation for localisation of the affected member. In the context of inspection planning, Varela and Tolentino (Varela and Tolentino, 2023) note that structural review intervals for fixed platforms can reach up to six years under certain conditions; this observation is used here to contextualise the maintenance horizon considered in the operational recommendations derived from this study.

4.1. Formulation of Vibratory Indicators

Following the framework of Ervin and Zeng (Ervin and Zeng, 2022), the detection system is fed by a set of damage indices (DIs) derived from the modal properties and flexibility matrix of the structure. Unlike the original work, which employs 51 uniaxial directional DIs and 24 triaxial resultant DIs, this study adopts eight scalar indicators representative of modal and flexibility changes, adapted to the three-dimensional jacket platform model. The mathematical formulations used as inputs to the GA are defined below, where u denotes the undamaged state, d the damaged state, N_m the number of vibration modes considered in the analysis (in this study, $N_m = 6$), and $\mathcal{R}_j(\cdot)$ the extraction operator that selects the scalar value corresponding to the j -th structural element from the argument vector.

4.1.1. Mode-Shape-Based Indicators

DI_1 : COMAC

COMAC quantifies the point-by-point correlation between two sets of mode shapes. The complement of its square root is employed for damage detection:

$$DI_1(j) = 1 - \sqrt{\frac{\left(\sum_{i=1}^{N_m} |\phi_{u,ij}| |\phi_{d,ij}|\right)^2}{\sum_{i=1}^{N_m} \phi_{u,ij}^2 \sum_{i=1}^{N_m} \phi_{d,ij}^2}} \quad (2)$$

DI_2 : Absolute Mode Shape Difference

Computes the magnitude of the change in the modal displacement vector:

$$DI_2(j) = \mathcal{R}_j \left(\sum_{i=1}^{N_m} |\phi_{u,i} - \phi_{d,i}| \right) \quad (3)$$

DI_3 : Relative Mode Shape Ratio

Evaluates the deviation with a regularisation threshold $\epsilon = 10^{-6}$ (to avoid numerical division by zero):

$$DI_3(j) = \mathcal{R}_j \left(\sum_{i=1}^{N_m} \left| \frac{\phi_{d,i}}{\phi_{u,i} + \epsilon} - 1 \right| \right) \quad (4)$$

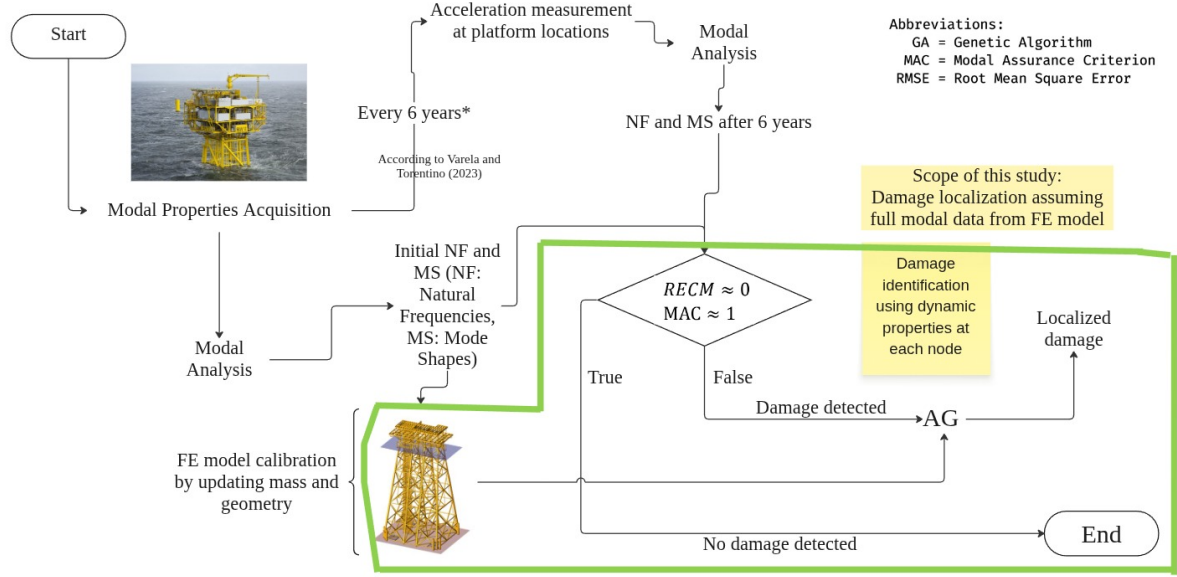


Figure 4: General scheme of the **proposed** damage identification methodology. The delimited box indicates the scope of the present work: numerical simulation of damage scenarios, extraction of **vibratory** indicators (DIs), and DQI+GA optimisation for localisation of the damaged member.

4.1.2. Flexibility-Based Indicators

The modal flexibility matrix is approximated as $\mathbf{F} \approx \Phi \mathbf{\Omega}^{-2} \Phi^T$.

DI_4 : Flexibility Difference

$$DI_4(j) = \mathcal{R}_j (|\text{diag}(\mathbf{F}_d) - \text{diag}(\mathbf{F}_u)|) \quad (5)$$

DI_5 : Flexibility Ratio

$$DI_5(j) = \mathcal{R}_j \left(\left| \frac{\text{diag}(\mathbf{F}_d)}{\text{diag}(\mathbf{F}_u) + \epsilon} - 1 \right| \right) \quad (6)$$

DI_6 : Percentage Flexibility Variation (PercF)

$$DI_6(j) = \mathcal{R}_j \left(100 \cdot \frac{|\text{diag}(\mathbf{F}_d) - \text{diag}(\mathbf{F}_u)|}{\text{diag}(\mathbf{F}_u) + \epsilon} \right) \quad (7)$$

4.1.3. Statistical Indicators

DI_7 : Flexibility Z-Score

Standardises the flexibility differences (ΔF_j):

$$DI_7(j) = \frac{\Delta F_j - \mu_{\Delta F}}{\sigma_{\Delta F}} \quad (8)$$

DI_8 : Damage Probability

Transforms the Z-score into a probability under a Gaussian distribution assumption:

$$DI_8(j) = 1 - 2 (1 - \Phi_{\text{cdf}}(|DI_7(j)|)) \quad (9)$$

4.2. The Search Engine: Genetic Algorithm (GA)

Following the principle of Ervin and Zeng (Ervin and Zeng, 2022), the GA acts as the optimisation engine to find

the optimal linear combination $\alpha = [\alpha_1, \dots, \alpha_8]$ of the eight DIs defined in the previous section. The differential contribution of this work lies in the fact that, unlike the binary target vector used by Ervin and Zeng, the fitness function evaluated here is the DQI (Equation 10), which directly penalises false positives and is calibrated for the offshore operational context. The computational architecture designed for this purpose is presented in Figure 5.

4.2.1. Genetic Operator Configuration

To ensure a balance between exploration of the solution space and exploitation of the best results, the following evolutionary parameters were defined:

- **Encoding (Chromosome):** Each individual in the population is represented as a real-valued vector $\alpha = [\alpha_1, \alpha_2, \dots, \alpha_8]$, normalised to the range $[0, 1]$.
- **Initial Population:** A population of $N = 300$ individuals is randomly initialised to ensure genetic diversity.
- **Selection and Crossover:** The Tournament method is used for parent selection, and a uniform arithmetic crossover operator generates offspring.
- **Mutation:** A Gaussian perturbation is applied to maintain diversity and avoid local optima.
- **Stopping Criterion:** The evolutionary process terminates upon reaching a **maximum of 200 generations** or if the mean fitness function shows no improvement exceeding 10^{-5} over **20 consecutive generations** (stagnation).

El ciclo detallado de estos operadores se ilustra en la Figura 6.

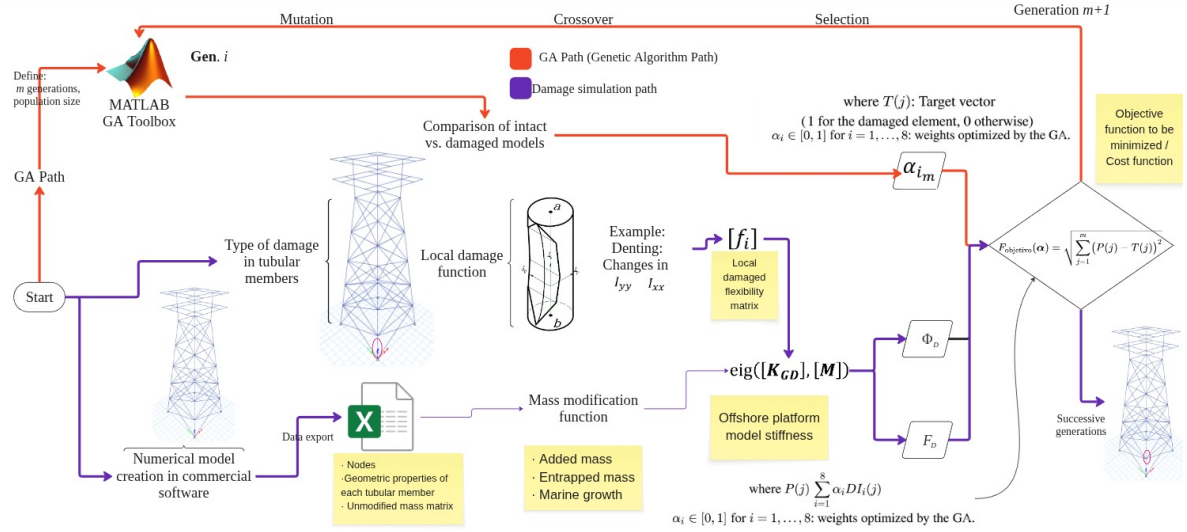


Figure 5: Computational architecture of the GA configured for data fusion.

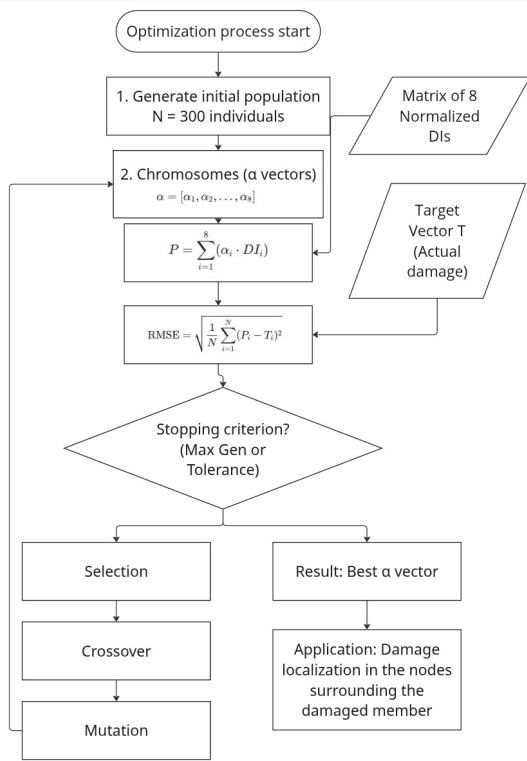


Figure 6: Detail of the evolutionary cycle and GA operators.

4.3. Objective Function: Detection Quality Index (DQI)

The evaluation of performance in offshore structures requires robust metrics. In this work the **Detection Quality Index (DQI)** is proposed, a composite metric designed to validate the operational utility of the system.

The DQI is defined mathematically as:

$$DQI = D \times C_{\text{norm}}(\delta) \times P_{\text{FP}}(N_{\text{FP}}) \quad (10)$$

where the following factors are integrated:

4.3.1. Localisation Success Factor (D)

Evaluates spatial accuracy. A value of $D = 1.0$ is assigned for exact detection of the target element, $D = 0.5$ for detection in immediately adjacent elements (acknowledging stress redistribution in lattice structures), and $D = 0.0$ for localisation failure.

4.3.2. Logarithmic Confidence Factor (C_{norm})

This component acts as a **Confidence Weighting Factor**. Given the logarithmic nature of the formulation:

$$C_{\text{norm}}(\delta) = \frac{\ln(1 + \gamma \cdot \delta)}{\ln(1 + \gamma \cdot \delta_{\text{max}})} \quad (11)$$

the index assigns a higher score to the detection of severe damage, where the **vibratory** signal is clear and the certainty level is high. Conversely, it penalises incipient damage cases (low severity δ), where the signal-to-noise ratio is poor and detection entails greater inherent uncertainty. It is therefore a conservative index that prioritises certainty in structural diagnosis.

4.3.3. False-Positive Penalisation Factor (P_{FP})

False alarms are exponentially penalised through $P_{\text{FP}} = e^{-\beta \cdot N_{\text{FP}}}$, with $\beta = 0.15$. This reflects the operational reality in which the accumulation of false positives rapidly degrades confidence in the monitoring system.

5. Results and Discussion

This section presents the findings obtained by applying the data fusion methodology based on the Detection Quality Index (DQI). Validation was conducted on the numerical jacket platform model, subjecting it to progressive corrosion

(uniform thickness reduction) and denting (localised geometric damage) scenarios. The analysis focuses on the sensitivity of the index in discriminating damage severity and, crucially, its capacity to distinguish between structurally critical members (legs) and redundant or “fuse” members (diagonals).

5.1. Sensitivity to Corrosion Scenarios

Corrosion was simulated as a uniform cross-sectional loss. Figure 7 illustrates the evolution of the DQI as a function of damage percentage, broken down by element type: beams, braces, and legs.

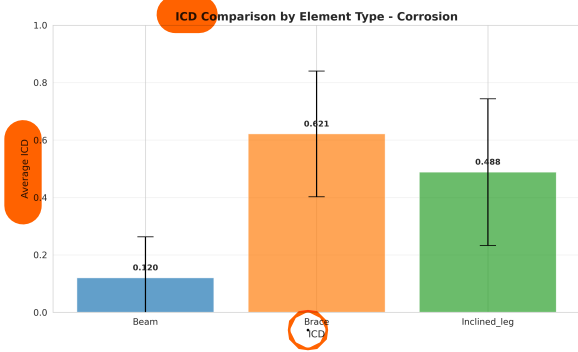


Figure 7: DQI sensitivity curves by element type under progressive corrosion scenarios.

A clear dichotomy is observed in the system response. Secondary elements (braces and inclined members) exhibit a high DQI growth rate, crossing the reliable detection threshold ($DQI > 0.7$) at damage levels as low as 30%. This validates the algorithm’s capacity to exploit local mode sensitivity in lower-stiffness members.

In contrast, main legs display a nearly flat response curve until cross-sectional loss exceeds 50–60%. This behaviour is attributed to the high axial and bending stiffness of the legs, which dominates the global structural response; small perturbations in these members are masked by structural redundancy, hampering early identification through frequency or global mode shape changes.

5.2. Sensitivity to Denting Scenarios

Figure 8 presents the results for impact damage (denting), which produces a drastic but localised reduction of the moment of inertia.

Although the physical nature of the damage differs from corrosion, the detectability trend is maintained: diagonal members act as early anomaly indicators. It is noteworthy that, for denting, the detection slope for legs is even more critical, requiring severe deformation before the GA can isolate them. This suggests that, for the monitoring of main legs, the DQI should be supplemented with local inspections in high impact-risk zones.

5.3. Zone Influence Analysis (Depth)

To determine whether depth affects diagnostic reliability, the structure was segmented into the five vertical zones

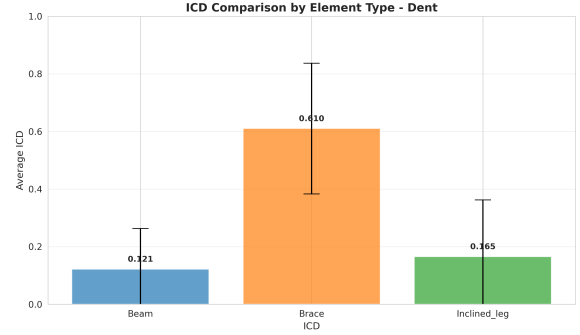


Figure 8: Comparison of DQI sensitivity by element type under denting scenarios.

defined in Figure 10: the **Splash Zone**, three intermediate submerged spans (Sub 1, Sub 2, and Sub 3), and the **Mudline**. Figures 9a and 9b compare DQI performance across these regions.

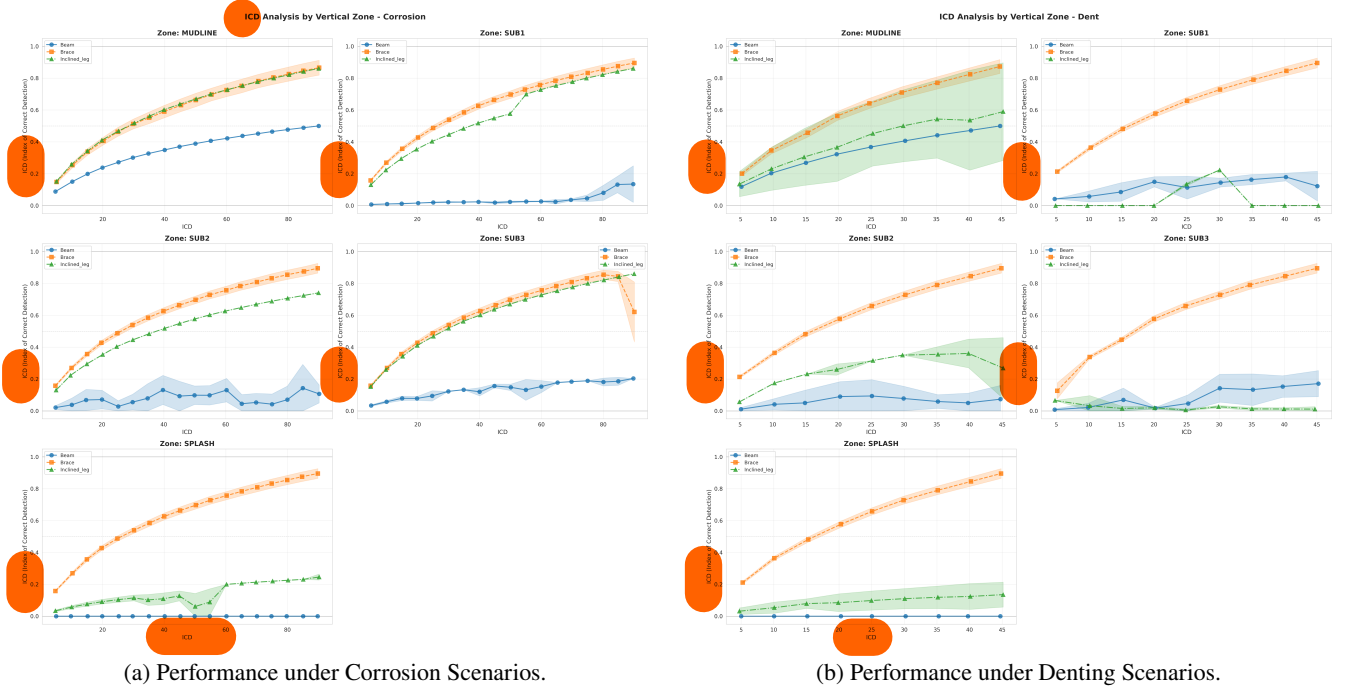


Figure 9: Influence of vertical damage location on detection quality (DQI).

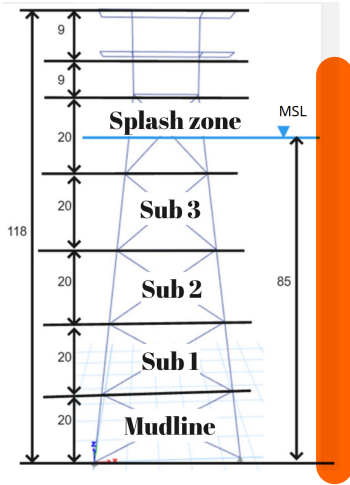


Figure 10: Vertical zonation of the jacket-type platform: depth-based delineation of the study zones. Elevations are expressed in metres referenced to Mean Sea Level (MSL).

The analysis reveals a slight advantage in the detectability of damage located in the **Splash Zone and Mid-Zone** compared to the **Mudline**. This is consistent with modal theory: fundamental bending modes display larger displacement and curvature amplitudes at the top of the structure (cantilever behaviour), improving the signal-to-noise ratio for the identification algorithms. Nevertheless, the stiffness hierarchy (Element Type) remains the predominant factor over the spatial location of damage.

5.4. Experimental Design and Case Coverage

Statistical validation is supported by a total of 3,240 GA runs: 1,080 for denting (120 elements $\times$ 9 severity levels: 5%–45%) and 2,160 for corrosion (120 elements $\times$ 18 severity levels: 5%–90%). Table 1 summarises the distribution of runs by damage type and element type.

Figure 11 summarises the experimental design from three complementary perspectives. Panel (a) confirms the intentional asymmetry in the number of runs: 2,160 for corrosion versus 1,080 for denting, a difference that reflects the wider severity range explored (18 vs. 9 damage levels). Panel (b) shows that brace elements (X-diagonals) constitute **67%** of the total 120 structural members (80 elements), a proportion that confers greater statistical representativeness to the results for this element type—precisely those showing the best detection performance in the subsequent analysis. Panel (c) confirms that each vertical level of the model contains an equivalent typological composition, ensuring inter-zone comparability.

The platform is divided into five vertical levels (see Figure 10): the **Mudline** (base) and three intermediate submerged spans (Sub 1, Sub 2, and Sub 3) of 20 m each, plus the **Splash Zone** at the air–water interface, resulting in a total submerged draft of 85 m referenced to Mean Sea Level.

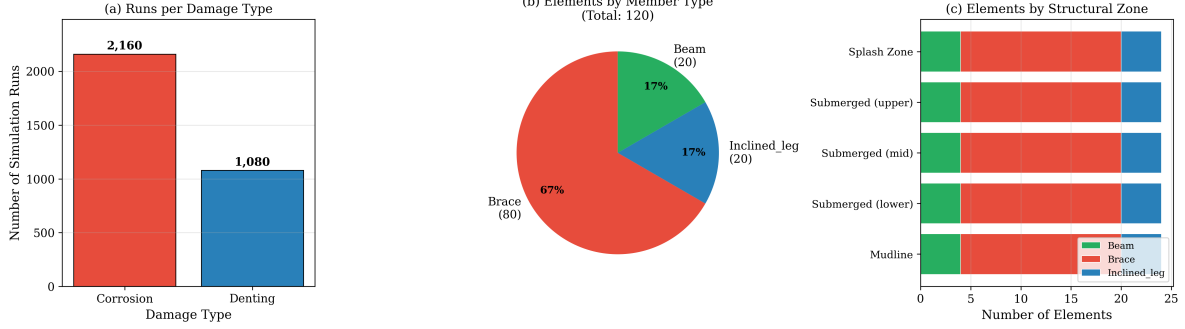
5.5. Classification Metrics: Precision and Recall

To quantify system performance beyond the DQI, standard binary classification metrics are employed, grounded in the **confusion matrix** of the problem. In each GA run, the algorithm produces a prediction vector $\hat{\mathbf{d}} \in \{0, 1\}^{N_{el}}$ (1 = element predicted as damaged) compared against the ground-truth label $\mathbf{d}^* \in \{0, 1\}^{N_{el}}$ (exactly one damaged

Table 1

Summary of the numerical experiment: damage scenarios evaluated with the DQI+GA framework.

Damage Type	N Elements	Severity Range [%]	Step [%]	Levels	Total Runs
Denting	120	5–45	5	9	1,080
Corrosion	120	5–90	5	18	2,160
Total					3,240

Experimental Design: Damage Scenarios Distribution**Figure 11:** Experimental design: (a) simulation runs by damage type (Corrosion: 2,160; Denting: 1,080); (b) typological composition of the 120 model elements (*Braces*: 67%, *Beams*: 17%, *Inclined legs*: 17%); (c) element distribution by vertical structural zone.

element per run). Four outcomes arise from this comparison: *True Positive* (TP: damaged element correctly identified), *False Positive* (FP: healthy element flagged as damaged), *False Negative* (FN: damaged element not detected), and *True Negative* (TN: healthy element correctly discarded). The two principal derived metrics are:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (12)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (13)$$

In the structural integrity context, Precision quantifies how many of the elements flagged as damaged are truly damaged (false inspection alarm cost); Recall quantifies how many truly damaged elements are detected (missed-damage cost). Table 2 reports the mean values by element type and damage scenario.

Table 2 shows that Recall values are consistently high (close to 1.0), confirming the DQI's capacity to detect the damaged element. However, Precision values are systematically low—particularly for Beams (0.08–0.12)—as a consequence of an inherent limitation of the classical metric: it penalises all False Positives (FPs) identically, irrespective of their topological distance to the damaged element. In the jacket lattice network, an adjacent FP ($d = 1$) reflects stress redistribution in the immediate vicinity and carries a radically different engineering meaning from one at the far end of the structure ($d = 8$). This uniform penalisation systematically underestimates the true localisation quality of the algorithm and reduces the statistical value of Precision as a diagnostic reliability indicator in lattice

platforms. Section 5.6 introduces the **Topology-Weighted Precision** (Precision_w), which corrects this deficiency by assigning weights proportional to the topological distance of each FP, providing a more robust assessment of localiser performance.

5.6. Topology-Weighted Precision

5.6.1. Limitation of Classical Precision

Classical Precision treats all False Positives (FPs) with equal penalisation, regardless of their spatial proximity to the truly damaged element:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (14)$$

This formulation is inadequate in lattice systems where an FP node adjacent to the damaged element (topological distance $d = 1$) carries a very different engineering interpretation from one located at the opposite end of the structure ($d = 8$).

5.6.2. Topology-Weighted Precision Formulation

A novel metric is proposed that assigns a weight $w(d) \in [0, 1]$ to each FP as a function of its topological distance d to the damaged element, measured in number of structural members via **breadth-first search** (BFS) on the connectivity graph:

$$w(d) = \frac{d}{d + 1} \quad (15)$$

Table 2

Summary of classification performance at key severity levels. The low Precision values reflect a structural limitation of the classical metric, which assigns equal penalty to all false positives regardless of their topological distance to the damaged element (see Section 5.6).

Damage Type	Element	Severity	Precision	Recall
Denting	Brace	10%	0.404	0.952
Denting	Brace	20%	0.398	0.952
Denting	Brace	30%	0.365	0.952
Denting	Brace	45%	0.370	0.952
Denting	Inclined Leg	10%	0.152	0.600
Denting	Inclined Leg	20%	0.093	0.750
Denting	Inclined Leg	30%	0.122	1.000
Denting	Inclined Leg	45%	0.078	0.700
Denting	Beam	10%	0.084	1.000
Denting	Beam	20%	0.094	1.000
Denting	Beam	30%	0.116	1.000
Denting	Beam	45%	0.098	1.000
Corrosion	Brace	20%	0.381	0.952
Corrosion	Brace	40%	0.377	0.952
Corrosion	Brace	60%	0.381	0.952
Corrosion	Brace	90%	0.319	0.905
Corrosion	Inclined Leg	20%	0.286	1.000
Corrosion	Inclined Leg	40%	0.263	1.000
Corrosion	Inclined Leg	60%	0.333	1.000
Corrosion	Inclined Leg	90%	0.339	1.000
Corrosion	Beam	20%	0.099	1.000
Corrosion	Beam	40%	0.105	1.000
Corrosion	Beam	60%	0.105	1.000
Corrosion	Beam	90%	0.121	1.000

so that adjacent nodes ($d = 1$) receive only 50% of the penalisation, while distant nodes ($d \rightarrow \infty$) approach the full penalty. The **Topology-Weighted Precision** is then:

$$\text{Precision}_w = \frac{TP}{TP + \sum_{i \in FP} w(d_i)} \quad (16)$$

5.6.3. FP Distance Distribution

Figure 12 reveals that approximately 40% of all FP nodes are directly adjacent to the damaged element ($d = 1$), with a mean distance of $\bar{d} \approx 2.2$ members. This finding justifies the need for the topological correction: classical Precision systematically underestimates the localisation quality of the GA.

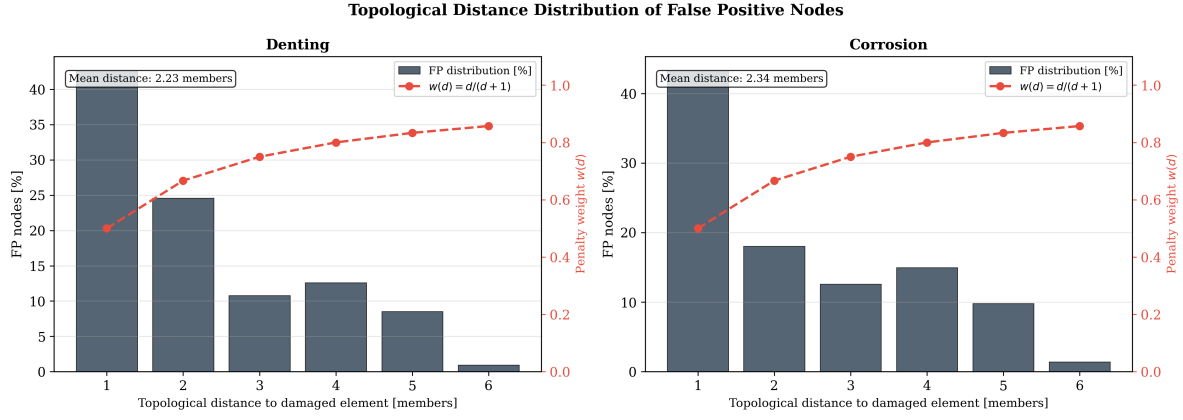


Figure 12: Topological distance distribution of FP nodes to the damaged element for Denting (left) and Corrosion (right). The red curve shows the weight function $w(d) = d/(d + 1)$.

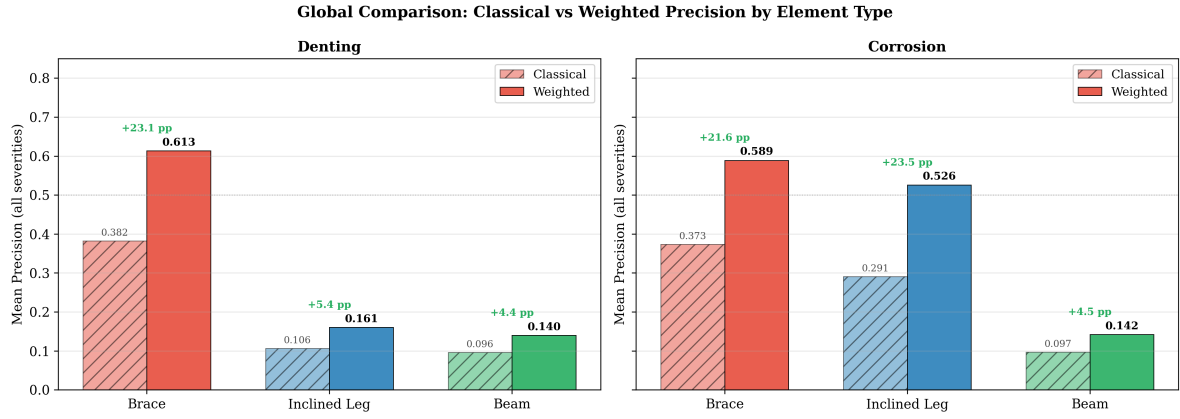


Figure 13: Global comparison of Classical vs. Topology-Weighted Precision by element type.

5.6.4. Quantitative Comparison

Figure 13 quantifies the Precision improvement for each element type when transitioning from the classical to the weighted metric. Braces show the greatest gain ($\Delta \approx +22$ percentage points) in both denting and corrosion scenarios, confirming that their FP signal is structurally coherent with the localised damage. Figure 14 details the evolution of classical and weighted Precision as a function of damage severity.

Figures 13 and 14 are presented below: the first summarises the global comparison by element type and the second details the evolution by severity level.

Classical Precision vs Topology-Weighted Precision

$$\text{Precision}_w = TP / (TP + \sum_i w(d_i)), \quad w(d) = d/(d+1)$$

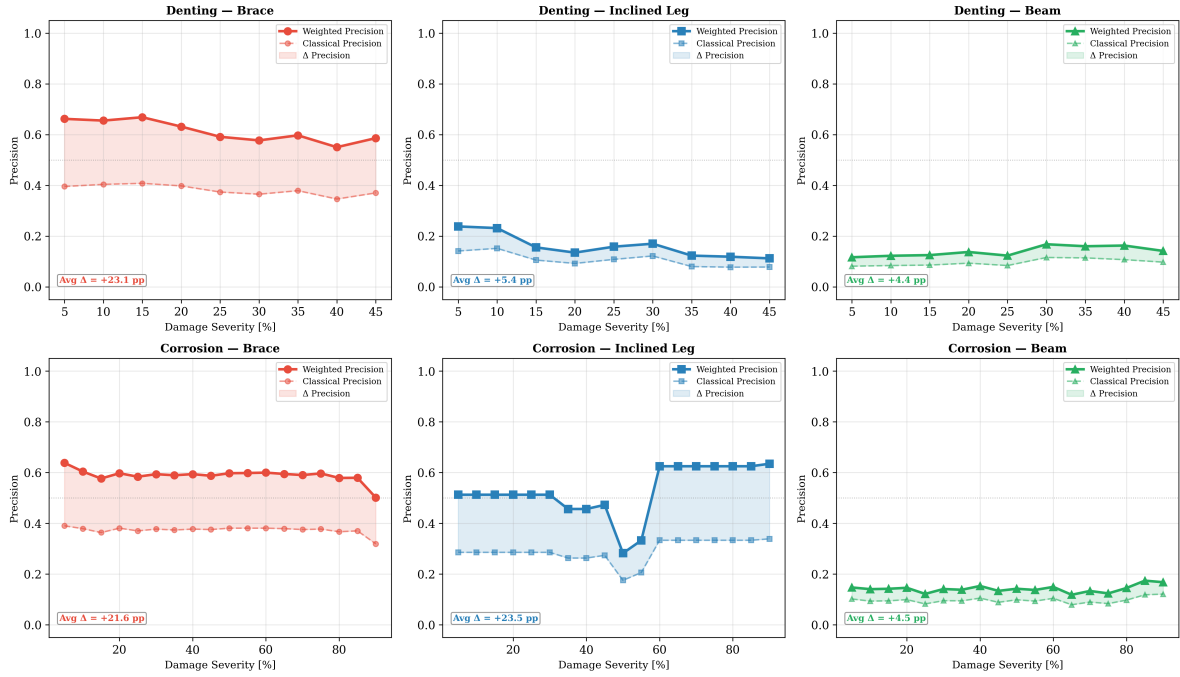


Figure 14: Evolution of Classical Precision (dashed line) and Topology-Weighted Precision (solid line) as a function of damage severity.

ICD Robustness: AUC-ROC Degradation vs. Noise Level (SNR)

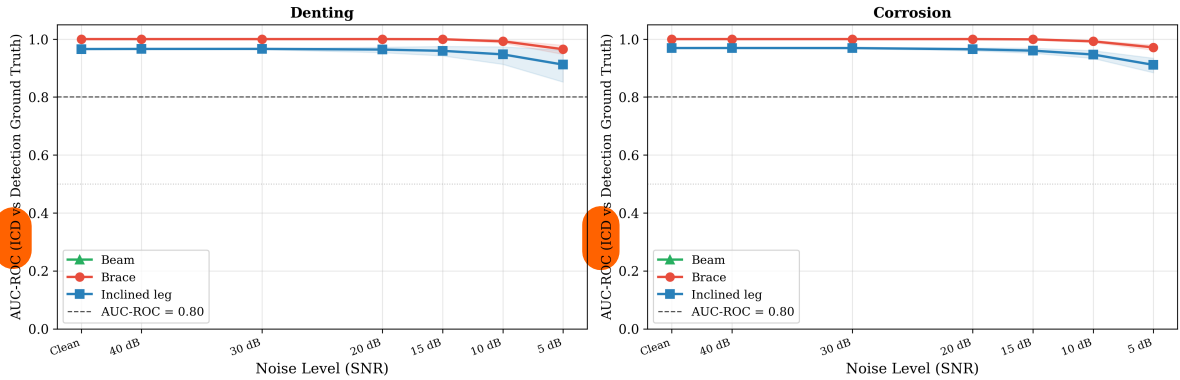


Figure 15: AUC-ROC degradation as a function of noise level (SNR) for Denting (left) and Corrosion (right). Dashed line = AUC = 0.80 threshold.

Table 4 (Appendix A) reports the complete numerical values by scenario, element type, and severity, including the mean FP distance.

5.7. Robustness to Gaussian Noise (SNR)

To evaluate the applicability of the method under realistic measurement conditions, controlled additive Gaussian noise with SNR levels between 5 and 50 dB was injected into the modal vectors prior to DQI evaluation. Figure 15 shows the AUC-ROC degradation as a function of SNR level for the three element types.

The results demonstrate that the DQI maintains AUC-ROC > 0.80 up to noise levels of SNR = 20 dB for braces, corresponding to moderately noisy measurement conditions typical of MEMS systems installed on operating offshore platforms. Table 3 summarises the AUC-ROC values at key SNR points.

Table 3

AUC-ROC of the DQI framework under additive white Gaussian noise (AWGN) at decreasing SNR levels. Ground truth: strict criterion ($\text{DetectionOK}=1.0$). Monte Carlo estimation with $N_{MC} = 500$ repetitions per SNR level.

Damage Type	Member Type	AUC-ROC at SNR level				Degradation Clean→5 dB
		Clean	20 dB	10 dB	5 dB	
Denting	Beam	N/A*	N/A*	N/A*	N/A*	N/A*
	Brace	1.000	1.000	0.992	0.965	+3.5%
	Inclined leg	0.966	0.964	0.947	0.912	+5.6%
Corrosion	Beam	N/A*	N/A*	N/A*	N/A*	N/A*
	Brace	1.000	1.000	0.992	0.971	+2.9%
	Inclined leg	0.969	0.965	0.947	0.911	+6.0%

SNR: Signal-to-Noise Ratio in decibels. AUC-ROC: Area Under the Receiver Operating Characteristic curve (range [0.5, 1.0], higher is better). Degradation (%): relative AUC-ROC loss from clean to 5 dB SNR. * N/A: AUC-ROC is undefined for Beam members because the DQI never achieves correct element localisation ($\text{DetectionOK}=0.5$ in all Beam cases, i.e., detection occurs but is attributed to the wrong element). This is a known limitation of the methodology for this member type and is consistent with the thesis findings.

6. Conclusions and Future Work

This study has validated the effectiveness of the Detection Quality Index (DQI) as a robust metric for structural damage identification in jacket-type offshore platforms. The adopted approach extends the GA-based DI combination framework introduced by Ervin and Zeng (Ervin and Zeng, 2022) to the offshore jacket domain, incorporating the DQI as the fitness function. Unlike the original binary target vector, the DQI exponentially penalises false positives and weights damage severity. This extension, together with the statistical validation over 3,240 scenarios, constitutes the central methodological contribution of the work. The campaign of 3,240 runs (1,080 denting + 2,160 corrosion) on 120 ETABS model elements provides exhaustive statistical support for the reported results.

The most relevant finding confirms the behaviour of secondary members (diagonals or braces) as *structural fuses*. The results demonstrate that the DQI is highly sensitive to damage in these elements, detecting stiffness reductions from 30% severity (for both corrosion and denting) with global detection rates of **92.2%** (denting) and **96.5%** (corrosion), and AUC-ROC values exceeding 0.90 for all member types. By contrast, primary members (legs) exhibited significant redundancy, requiring damage levels above 50% to become detectable by the system. This has a direct operational implication for offshore integrity management: while vibration-based automated monitoring is well-suited to the continuous surveillance of braces, underwater visual inspections should continue to prioritise main nodes and legs, where modal changes are more subtle in early stages.

A Gaussian noise robustness analysis demonstrates that the AUC-ROC remains above 0.80 up to an SNR level of 20 dB for braces, which is consistent with the measurement conditions of MEMS systems installed on operating platforms.

As a novel methodological contribution, the **Topology-Weighted Precision** (Precision_w) is proposed. This metric corrects the uniform penalisation of FPs by assigning a weight $w(d) = d/(d + 1)$ based on their distance in the

structural graph. The analysis shows that approximately 40% of FP nodes are adjacent to the damaged element ($d = 1$), implying that classical Precision systematically underestimates localisation quality. The new metric raises brace Precision by up to **+22 percentage points**, providing a fairer assessment of localiser performance in structural networks.

Furthermore, the DQI formulation proved effective in penalising false positives—a critical attribute for avoiding unnecessary alarms that increase operational costs in offshore facilities.

6.1. Future Work

To consolidate the industrial applicability of this method, the following research directions are proposed:

- **Experimental Validation:** Apply the methodology to laboratory-scale structures or real data from **structural health monitoring** (SHM) systems installed on operating platforms, with the aim of quantifying the effect of actual measurement uncertainties (experimental noise) versus simulated numerical noise, thereby validating the robustness of the DQI outside the idealised environment.
- **Environmental Uncertainty:** Incorporate stochastic variables simulating environmental noise (wave, wind) in the input signals, evaluating the robustness of the DQI against contaminated data.
- **Typological Extension:** Adapt the objective function and **vibratory** indicators for other offshore wind support typologies, such as tripods or monopiles, which exhibit modal dynamics different from jacket-type lattice structures.

A. Supplementary Tables

Table 4: Comparison of Classical and Topology-Weighted Precision. Δ = Weighted – Classical (percentage points). Avg FP dist: mean topological distance (in structural members) between false-positive nodes and the damaged element. Weight function: $w(d) = d/(d + 1)$.

Type	Element	Sev. [%]	Prec. Cl.	Prec. Wgt.	Δ [pp]	FP dist
Denting	Brace	5	0.396	0.662	+26.7	2.20
Denting	Brace	10	0.404	0.655	+25.1	2.16
Denting	Brace	15	0.408	0.669	+26.1	2.07
Denting	Brace	20	0.398	0.632	+23.4	4.40
Denting	Brace	25	0.374	0.592	+21.8	4.09
Denting	Brace	30	0.365	0.577	+21.2	4.01
Denting	Brace	35	0.379	0.597	+21.8	4.22
Denting	Brace	40	0.346	0.551	+20.4	3.66
Denting	Brace	45	0.370	0.586	+21.6	4.04
Denting	Inclined Leg	5	0.142	0.238	+9.7	1.51
Denting	Inclined Leg	10	0.152	0.232	+8.0	1.52
Denting	Inclined Leg	15	0.106	0.156	+5.0	1.71
Denting	Inclined Leg	20	0.093	0.135	+4.3	1.73
Denting	Inclined Leg	25	0.108	0.159	+5.0	1.59
Denting	Inclined Leg	30	0.122	0.170	+4.8	1.57
Denting	Inclined Leg	35	0.080	0.123	+4.3	1.70
Denting	Inclined Leg	40	0.078	0.119	+4.1	1.74
Denting	Inclined Leg	45	0.078	0.112	+3.4	1.79
Denting	Beam	5	0.082	0.117	+3.5	1.70
Denting	Beam	10	0.084	0.123	+3.8	1.71
Denting	Beam	15	0.086	0.125	+3.9	1.60
Denting	Beam	20	0.094	0.138	+4.4	1.47
Denting	Beam	25	0.084	0.123	+3.9	1.58
Denting	Beam	30	0.116	0.168	+5.2	1.45
Denting	Beam	35	0.114	0.160	+4.6	1.48
Denting	Beam	40	0.107	0.163	+5.6	1.38
Denting	Beam	45	0.098	0.142	+4.4	1.47
Corrosion	Brace	5	0.390	0.638	+24.8	2.89
Corrosion	Brace	10	0.379	0.604	+22.5	3.14
Corrosion	Brace	15	0.364	0.577	+21.3	3.13
Corrosion	Brace	20	0.381	0.597	+21.7	3.37
Corrosion	Brace	25	0.370	0.584	+21.3	3.24
Corrosion	Brace	30	0.377	0.594	+21.6	3.30
Corrosion	Brace	35	0.374	0.590	+21.6	3.32
Corrosion	Brace	40	0.377	0.594	+21.7	3.37
Corrosion	Brace	45	0.376	0.587	+21.2	2.89
Corrosion	Brace	50	0.381	0.598	+21.7	3.25
Corrosion	Brace	55	0.381	0.598	+21.7	3.26
Corrosion	Brace	60	0.381	0.600	+21.9	3.22
Corrosion	Brace	65	0.379	0.595	+21.6	3.27
Corrosion	Brace	70	0.376	0.590	+21.5	4.31
Corrosion	Brace	75	0.377	0.597	+21.9	4.21
Corrosion	Brace	80	0.367	0.579	+21.2	4.15
Corrosion	Brace	85	0.370	0.580	+20.9	3.68
Corrosion	Brace	90	0.319	0.502	+18.2	3.21
Corrosion	Inclined Leg	5	0.286	0.513	+22.7	1.29
Corrosion	Inclined Leg	10	0.286	0.513	+22.7	1.29
Corrosion	Inclined Leg	15	0.286	0.513	+22.7	1.29
Corrosion	Inclined Leg	20	0.286	0.513	+22.7	1.29

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Type	Element	Sev. [%]	Prec. Cl.	Prec. Wgt.	Δ [pp]	FP dist
Corrosion	Inclined Leg	25	0.286	0.513	+22.7	1.29
Corrosion	Inclined Leg	30	0.286	0.513	+22.7	1.29
Corrosion	Inclined Leg	35	0.263	0.457	+19.3	1.39
Corrosion	Inclined Leg	40	0.263	0.457	+19.3	1.39
Corrosion	Inclined Leg	45	0.274	0.473	+19.9	1.44
Corrosion	Inclined Leg	50	0.175	0.282	+10.7	1.38
Corrosion	Inclined Leg	55	0.206	0.333	+12.7	1.60
Corrosion	Inclined Leg	60	0.333	0.625	+29.2	1.00
Corrosion	Inclined Leg	65	0.333	0.625	+29.2	1.00
Corrosion	Inclined Leg	70	0.333	0.625	+29.2	1.00
Corrosion	Inclined Leg	75	0.333	0.625	+29.2	1.00
Corrosion	Inclined Leg	80	0.333	0.625	+29.2	1.00
Corrosion	Inclined Leg	85	0.333	0.625	+29.2	1.00
Corrosion	Inclined Leg	90	0.339	0.635	+29.6	1.00
Corrosion	Beam	5	0.102	0.147	+4.5	1.43
Corrosion	Beam	10	0.094	0.141	+4.7	1.38
Corrosion	Beam	15	0.095	0.142	+4.7	1.34
Corrosion	Beam	20	0.099	0.146	+4.7	1.38
Corrosion	Beam	25	0.082	0.122	+4.0	1.49
Corrosion	Beam	30	0.095	0.141	+4.6	1.45
Corrosion	Beam	35	0.095	0.138	+4.4	1.38
Corrosion	Beam	40	0.105	0.153	+4.8	1.38
Corrosion	Beam	45	0.089	0.134	+4.5	1.43
Corrosion	Beam	50	0.099	0.142	+4.4	1.47
Corrosion	Beam	55	0.094	0.138	+4.4	1.46
Corrosion	Beam	60	0.105	0.150	+4.5	1.44
Corrosion	Beam	65	0.079	0.119	+3.9	1.44
Corrosion	Beam	70	0.090	0.134	+4.4	1.45
Corrosion	Beam	75	0.084	0.124	+4.0	1.47
Corrosion	Beam	80	0.098	0.145	+4.8	1.40
Corrosion	Beam	85	0.119	0.174	+5.6	1.37
Corrosion	Beam	90	0.121	0.168	+4.7	1.49

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